

Prompteam at ArgMining 2026: RAG-Augmented Classification and Cosine-Filtered Relation Prediction for UN Resolutions

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Abstract

We describe our system for the UZH ArgMining 2026 Shared Task on reconstructing argumentative structure in UN/UNESCO resolutions. The task requires (1) classifying paragraph types and assigning thematic tags from a 141-label taxonomy, and (2) predicting directed argumentative relations between paragraphs. Our pipeline combines a quantised Qwen2.5-7B-Instruct model with retrieval-augmented generation (RAG) backed by FAISS-indexed dense embeddings for few-shot prompting and tag candidate pre-filtering. For relation prediction, we apply a sliding-window cosine pre-filter that reduces the quadratic pair space to near-linear cost. A parallelisable, fault-tolerant pipeline with atomic checkpointing enabled complete processing of 2,959 paragraphs across three concurrent Kaggle T4 sessions despite 12-hour GPU limits. Our system achieved **2nd place** overall on the shared task leaderboard.

1 Introduction

The UZH ArgMining 2026 Shared Task (Gao and others, 2026) targets the analysis of argumentative structure in UN/UNESCO resolutions. **Subtask 1** requires per-paragraph prediction of (a) a binary structural type (*preambular* vs. *operative*) and (b) a multi-label subset from a 141-label thematic taxonomy spanning 15 dimensions. **Subtask 2** requires identifying directed relations between paragraph pairs from four types: *supporting*, *contradictive*, *complemental*, and *modifying*. The final ranking averages an automated F1 score and an LLM-as-a-Judge score (0–100) that rates chain-of-thought reasoning quality. All teams must use open-weight models with $\leq 8B$ parameters.

Our contributions are:

- A RAG-based few-shot classification pipeline using FAISS-indexed dense embeddings for

tag candidate pre-filtering and dynamic example retrieval (§3.2).

- A cosine-similarity pre-filter reducing relation prediction from $O(N^2)$ to near-linear cost (§3.3).
- A fault-tolerant, parallelisable design with atomic checkpointing across Kaggle’s 12-hour GPU limit (§3.4).
- **2nd place** on the shared task leaderboard.

2 Related Work

Argument mining has progressed from essay-level parsing (Stab and Gurevych, 2017) and MST-based discourse prediction (Peldszus and Stede, 2015) to large-scale political text analysis (Lawrence and Reed, 2020). The UN-RES corpus (Gao and others, 2026) extends argument mining to multilingual UN resolutions with fine-grained thematic tagging. RAG pipelines (Lewis et al., 2020) ground LLM predictions in retrieved evidence; dynamically selecting semantically similar demonstrations outperforms random sampling in few-shot settings (Rubin et al., 2022). We apply this paradigm jointly to tag candidate retrieval and few-shot example selection, and reduce the quadratic cost of pairwise relation prediction via embedding-based cosine pre-filtering.

3 System Description

3.1 Data and Model Stack

The training set comprises 2,690 UN resolutions (Gao and others, 2026); the test set has 89 documents (45 UNESCO ICE texts, 1934–2008) with 2,959 paragraphs and a 141-label thematic taxonomy across 15 dimensions. Our LLM is Qwen2.5-7B-Instruct (Qwen Team, 2025) in INT8 quantisation (Dettmers et al., 2022) (~ 7 GB VRAM). Sentence embeddings use all-mpnet-base-v2 (768-d) (Reimers and

Gurevych, 2019) with FAISS IndexFlatIP (Johnson et al., 2021) for exact cosine search.

3.2 Subtask 1: Classification

Each paragraph passes through a four-stage pipeline (Figure 1): (1) **Embed** the paragraph into a 768-d vector; (2) **Tag retrieval** via cosine search returns the top-20 candidate tags; (3) **RAG few-shot** retrieves the 5 most similar annotated paragraphs as demonstrations; (4) **LLM classify** with a structured prompt yields a JSON with fields `think` (chain-of-thought), `type`, and `tags`. A heuristic fallback (first-word rule) applies if all three JSON-parse retries fail; output tags are sanitised against the taxonomy.



Figure 1: Subtask 1 pipeline per paragraph.

Prompt design. The system prompt defines the two paragraph types with canonical opening keywords, tag selection rules emphasising precision, and a mandatory think template with four reasoning steps (Wei et al., 2022): identify keyword $\rightarrow$ decide type $\rightarrow$ evaluate candidates $\rightarrow$ justify tags. Few-shot examples are ordered by descending similarity (Rubin et al., 2022).

3.3 Subtask 2: Relation Prediction

For each paragraph B , we determine which earlier paragraph A shares an argumentative link. Two strategies reduce the $O(N^2)$ cost (Figure 2):



Figure 2: Subtask 2 pipeline per paragraph pair.

Sliding window. Paragraph B is compared only against the preceding $w=8$ paragraphs, exploiting sequential rhetorical structure. **Cosine pre-filter.** Only pairs with cosine similarity $\geq \theta=0.30$ reach the LLM, eliminating $\sim 65\%$ of candidates. The prompt requests four-step reasoning: summarise $A \rightarrow$ summarise $B \rightarrow$ compare themes $\rightarrow$ assign relation(s).

3.4 Engineering Design

Parallel chunking. Both tasks distribute work across $N=3$ concurrent Kaggle notebooks via modulo indexing ($i \bmod N = K$), with a

merge step deduplicating by composite key (`doc_id||para_id`). Full hyperparameter settings are provided in Appendix A.

Atomic checkpointing. Results are written to a `.tmp` file and atomically renamed, preventing corruption on session termination. On resume, processed entries are skipped.

4 Results

Our system achieved **2nd place** on the ArgMining 2026 leaderboard.

4.1 Subtask 1: Classification

Metric	Value
Total paragraphs	2,959
Documents	89
Duplicates	0
Preambular / Operative	58.3% / 41.7%
Avg. tags per paragraph	2.4
Avg. think length	74 words

Table 1: Subtask 1 coverage and distribution.

Table 1 shows full coverage of all 2,959 paragraphs with zero duplicates after the composite-key fix. The type split (58.3% preambular) is consistent with UN resolution structure. The average of 2.4 tags per paragraph reflects focused multi-label predictions; the 74-word think field provides sufficient reasoning depth for LLM-as-a-Judge scoring.

4.2 Subtask 2: Relations

Relation Type	Share
Supporting	44%
Complemental	31%
Modifying	18%
Contradictive	7%
Pairs eliminated by pre-filter	$\sim 65\%$

Table 2: Subtask 2 relation distribution.

Table 2 shows that *supporting* relations dominate (44%), as expected in consensus-driven UN documents. *Contradictive* relations are rare (7%). The cosine pre-filter eliminated $\sim 65\%$ of candidate pairs, substantially reducing LLM inference cost.

4.3 Error Analysis

Two recurring failure modes were identified. First, with only $k=20$ tag candidates pre-filtered per paragraph, rare labels (e.g., INFRA_WASH) may never

enter the candidate set. Second, the cosine threshold occasionally filters out argumentatively related pairs with dissimilar vocabulary, particularly for *contradictive* relations.

5 Conclusion

We presented a complete, fault-tolerant pipeline for the UZH ArgMining 2026 Shared Task, achieving 2nd place by combining a quantised 7B instruction-tuned model with FAISS-backed retrieval-augmented generation and cosine-pre-filtered relation prediction. Key takeaways: (i) composite keys are essential for paragraph-level check-pointing across multi-document corpora; (ii) cosine pre-filtering reduces relation prediction cost to near-linear without substantial recall loss; (iii) explicit multi-step think prompts produce structurally richer reasoning traces for LLM-as-a-Judge evaluation. Future work includes fine-tuning a small model on the UN-RES training set, exploring cross-document relation links, and learning a calibrated cosine threshold via cross-validation.

Limitations

Tag recall. With 141 tags and only $k=20$ candidates pre-filtered by embedding similarity, rare thematic tags may be systematically missed. An ensemble of multiple embedding queries or a hierarchical tag-grouping strategy could improve recall.

Relation and reasoning quality. The cosine similarity threshold $\theta=0.30$ was set empirically without cross-validation, which may affect the precision–recall trade-off. Additionally, while prompts enforce structured reasoning, the quality of generated think chains remains uneven. Future work could incorporate learned thresholds or reranking models, along with improved reasoning elicitation techniques.

Reproducibility. Our pipeline depends on INT8 quantisation via bitsandbytes, which introduces non-deterministic rounding across hardware. As a result, outputs may vary slightly across GPU architectures.

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Appendix

A Hyperparameters

Table A lists the key hyperparameters used in our pipeline.

Component	Parameter	Value
LLM	Model	Qwen2.5-7B
	Quantisation	INT8
	Max new tokens	1,536
	Temperature	0.05
	Top- p	0.9
	Repetition penalty	1.1
	JSON retries	3
Embeddings	Model	mpnet-base-v2
	Dimensions	768
	Batch size	256
Subtask 1	RAG top- k	5
	Tag candidates	20
Subtask 2	Window size w	8
	Cosine threshold θ	0.30

Table A: Key hyperparameters.